EXPERIMENT EXECUTION ORDER

FinPersona-Bench | run sheet for all experiments on the new synthetic environment | 23 Aug 2026

Each step depends only on the steps above it. Do not start a step before the verification of the previous one is written down.

THE ORDER

	Step	What to run	Verify before moving on
0	**Freeze the environment**	Apply the remaining environment fixes; re-run the stylized-facts checklist, the leakage audit (L1, L2, L2b, L4) and the hazard calibration; run the L3 LLM probe (a few frontier models asked to estimate value/sign from the rendered observation); freeze the generator and record its config hash.	Checklist results recorded (passes and accepted fails, with reasons). L2 passes on held-out seeds (calm R2 of mispricing at or below threshold). L3 probe at or below the surrogate bound. Same seed always gives the same path. Config and prompt hashes stable.
1	**Harness smoke test**	One cheap model, a handful of runs per scenario and arm: static, memory, both placebos, wrapper-only, swapped, no-mandate trader, plus the restatement probe. Include one stateful run.	100 percent of calls parsed; every run has all rows; provenance columns (env version, config hash, prompt hash, temperature) filled; rendered prompt matches the published sample; the report and stats pipelines produce their tables without errors.
2	**M0: open-weight coverage and the t = 0 gate**	Full behavioural grid on the open-weight roster only (all personas, scenarios, seeds; static and memory arms; common-start slice for the gate). Compute band-MAS, mandate-conditional regret, drawdown and the day-1 separability gate per model.	Gate pass/fail recorded per model. Anchor models for the mechanistic work chosen from those that pass (or the failure is documented as a result). Zero-trade and fallback shares per model recorded.
3	**Main behavioural grid**	All models x 3 personas x all arms x all scenarios (crash per severity) x seeds x decode replicates, start-at-band-centre, Track B; plus the Track A slice, the O3 numerical-only arm, and the common-start slice.	Reliability table clean (parse and fallback rates, zero-trade shares). Baselines and oracles simulated per cell from the same start. No degenerate normalisations unflagged. Spot-check a few runs end to end against the logs.
4	**Causal decomposition analysis**	No new runs: analyse step 3. Contrasts vs static per arm; content test (swapped mandate moves allocation to the injected side), force test (directive placebo and wrapper-only vs mandate), sign-reversal by persona x regime.	Model-level statistics only (mixed effects, sign counts, BH across the metric family, effect sizes with CIs). The conclusion stated per model family, not pooled over runs.
5	**Persona and salience analysis**	Target-free salience surrogate per model per window (persona share vs directive share); OCEAN grid with trait-space sampling; Track A vs Track B comparison; O3 as ceiling.	Surrogate validated: label-permutation null near zero, O3 arm high. Blocked cross-validation by seed. Persona-share and directive-share reported with CIs.
6	**Stateful grid (the decay test)**	Stateful agent (real accumulating context, mandate in the system prompt) on the anchor models and scenarios; restatement probe on schedule; windowed statistics on adherence across the rollout.	Context actually grows (token counts logged per step); probe answers logged; decay tested against a constant-exposure null with phase controlled; the long-horizon control run for at least one model.
7	**Mechanistic: M1 and M2**	On the anchor models, replaying step 3 and step 6 prompts: attention to the mandate span (per arm, with placebo, uniform and first-token-sink controls; aligned by phase, not calendar day; stateful arm included). Extract the mandate-presence direction (contrastive prompts; swapped-mandate pairs) and the adherence direction on the common-start slice; validate projections against the behavioural metric.	Replay reproduces logged prompt hashes exactly. Action read-out matches real generation on a sample. Direction AUROC reported on held-out seeds with the input-readable confound checked. Attention series per phase, with CIs.
8	**Mechanistic: M3, then M4**	M3: steer the mandate-presence direction in real rollouts (dose-response, ablate-and-restore), scored with band-MAS, mandate-conditional regret and drawdown, on at least 3 seeds per model, including the sustained-bull control. M4 only after its gate: re-check which models show the crash reversal under the new action interface; attribute the decision logits only on a model that does.	M3: monotone dose-response inside the operating range; ablate worsens and restore recovers adherence; sign pattern by regime reported per model with CIs; sustained-bull behaves as predicted (adherence moves, regret does not). M4: gate table recorded first; the numeric-action logit readout validated before attribution.
9	**Controller and frequency sweep**	Injection-frequency sweep scored on the behavioural metric; then the adaptive controller: re-inject only when the M2 projection crosses a threshold; compare against	Controller beats or matches always-on where re-grounding helps and avoids the harm where it hurts; token cost reported on the same plot; results

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		always-on, the sweep and a score-based trigger, per regime, with token cost.	per regime, not pooled.
10	**Sensitivities**	Harness factors on a subset: cost tier and visibility, execution timing, start design, action interface, field order, horizon disclosure, mandate wording; generator sensitivities already scripted (engine variants, panic multiplier).	Each sensitivity changes or does not change the headline contrasts; state which conclusions are robust and which are factor-dependent.
11	**Final scoring, statistics and figures**	Re-score everything with the final metric definitions; assemble the mixed-effects tables, the audit tables and the figures; write the results sections from these tables only.	Every number in the paper traces to a run file and a script; every figure regenerated from code; statistics at model level with BH correction; the audit and checklist tables included in the paper.

RULES THAT APPLY AT EVERY STEP

1	Do not start a step before the verification of the previous step is written down (a short note or a generated table is enough).
2	Anything that changes the environment or the prompts after step 0 restarts from step 0: re-freeze, re-audit, new hashes.
3	All behavioural claims are made at model level; runs are replicates, not observations.
4	Mechanistic work (steps 7 to 9) runs only on models chosen in step 2 and only on replayed prompts whose hashes match the logs.
5	If a step produces a null, record it and continue; the design does not depend on any single result.

DEPENDENCIES AT A GLANCE

	Step	Needs	Feeds
0	**Freeze**	the built environment	everything
1	**Smoke**	0	2, 3
2	**M0 + gate**	1	anchor models for 7, 8, 9; headline gate table
3	**Main grid**	1	4, 5, 7; all behavioural headlines
4	**Causal analysis**	3	the content-vs-force conclusion; paper thesis
5	**Salience / OCEAN**	3	persona-share result
6	**Stateful grid**	1 (can run parallel to 3)	the decay result; 7 stateful replay
7	**M1 + M2**	2, 3, 6	8, 9
8	**M3 + M4**	7	the mechanism result
9	**Controller + sweep**	7 (M2)	the method result
10	**Sensitivities**	3	robustness statements
11	**Final scoring**	all	the paper